**UC PRACTICAL 2 JOURNAL:** Design and develop chat messaging app which is a location-based

**AIM**

Design and develop a location-based application that retrieves device coordinates and evaluates geographic boundaries (Geofencing).

**THEORETICAL OVERVIEW**

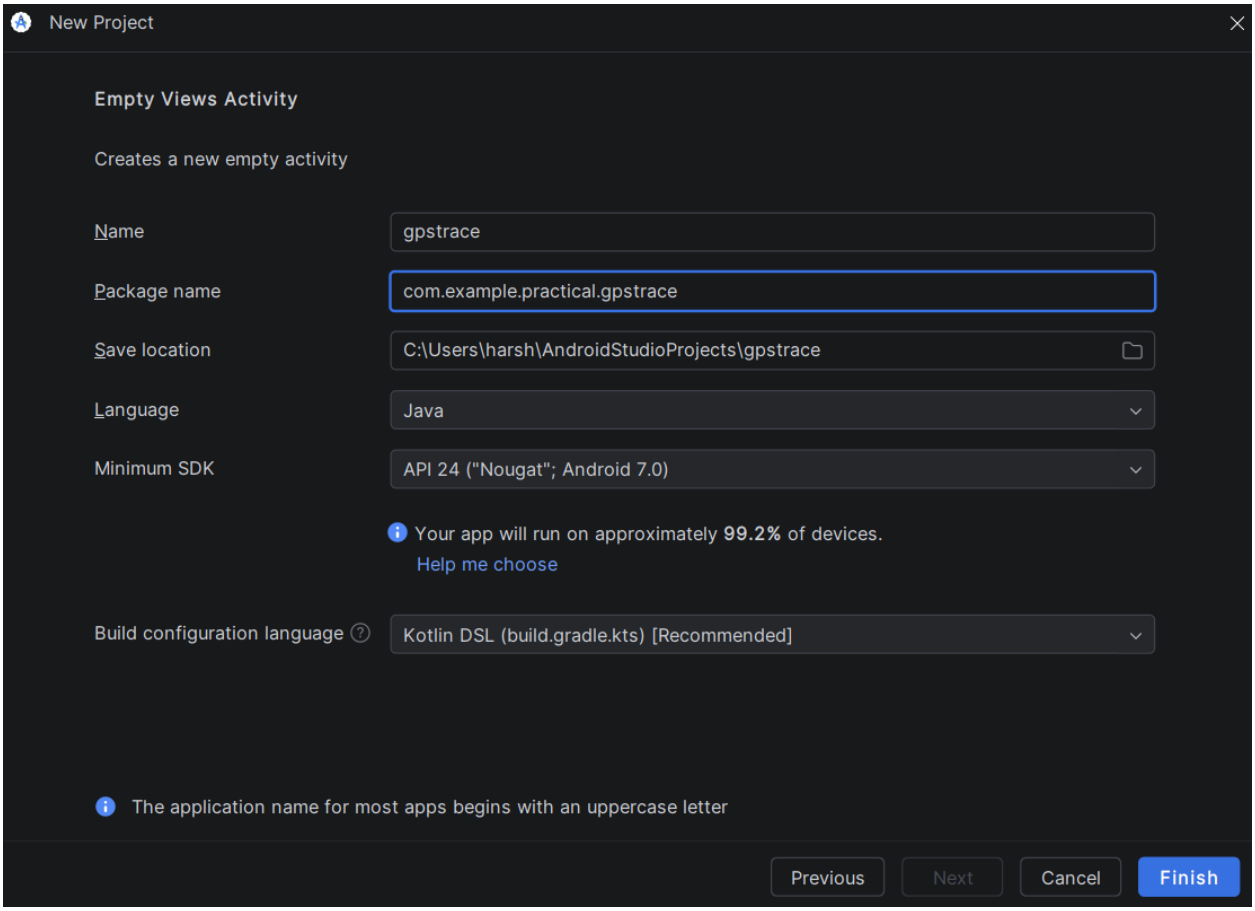
- **Location-Based Services (LBS):** Applications that leverage spatial metadata (Latitude and Longitude) provided by the device's hardware sensors to adapt functionality based on user context.
- **Dual Location Providers:**
  GPS_PROVIDER: Uses hardware satellites to deliver precise positioning (requires line-of-sight view outdoors).
  NETWORK_PROVIDER: Uses Wi-Fi networks and cell towers to triangulate positioning (works indoors and consumes less power).
- **Geofencing / Spatial Bounding Box:** Algorithmic evaluation of physical boundaries using logical bounds:
  if (currentLat > 36.0 && currentLat < 38.0 && currentLong > -122.0 && currentLong < -118.0)
- **Runtime Permissions Model:** Android API 23+ requires explicit dynamic security checks (ActivityCompat.requestPermissions) alongside manifest declarations before accessing location hardware.

**HARDWARE & SOFTWARE REQUIREMENTS**

- **IDE:** Android Studio (Quail Edition)
- **Language:** Java, XML
- **Testing Device:** Physical Android Device (e.g., Oppo Reno 10x Zoom, Android 12) or AVD Emulator
- **Connection:** USB Debugging Interface

**PROCEDURE / PRACTICAL STEPS**

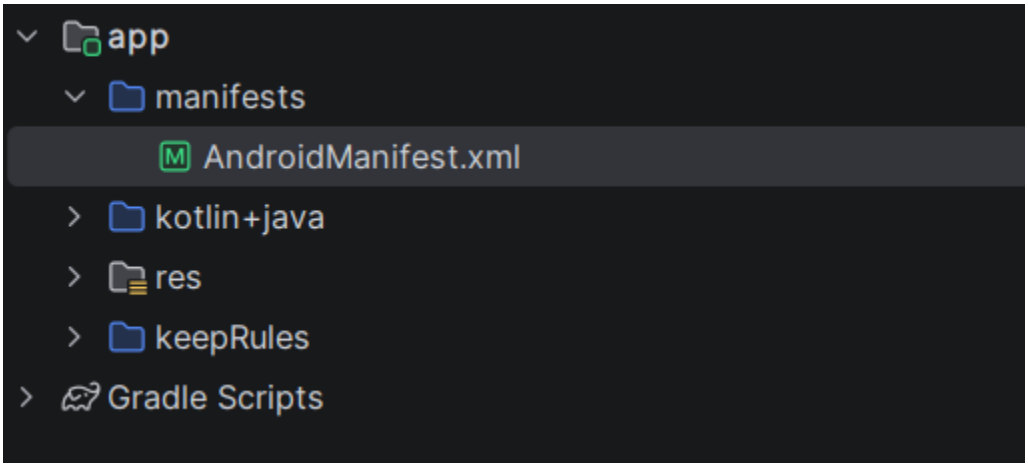
1. **Project Setup:**
   - Open Android Studio, create a new **Empty Views Activity** project.
   - Name the project gpstrace and set the package name to com.example.practical.gpstrace.




2. **Configuration & Design:**
    - Update AndroidManifest.xml with location access permissions.
    - Define the layout structure in activity_main.xml using RelativeLayout.
    - Implement logic in MainActivity.java to fetch cached location telemetry, register location updates silently, and render UI context on button click.

**SOURCE CODE**

# AndroidManifest.xml



```xml
<?xml version="1.0" encoding="utf-8"?>

<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    package="com.example.practical.gpstrace">

    <!-- Location Permissions -->
    <uses-permission
        android:name="android.permission.ACCESS_FINE_LOCATION" />

    <uses-permission
        android:name="android.permission.ACCESS_COARSE_LOCATION" />

    <!-- Internet Permission -->
    <uses-permission
        android:name="android.permission.INTERNET" />

    <application
        android:allowBackup="true"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:supportsRtl="true"
        android:theme="@style/Theme.AppCompat.Light.NoActionBar">

        <activity
            android:name=".MainActivity"
            android:exported="true">

            <intent-filter>

                <action
                    android:name="android.intent.action.MAIN" />

                <category
                    android:name="android.intent.category.LAUNCHER" />

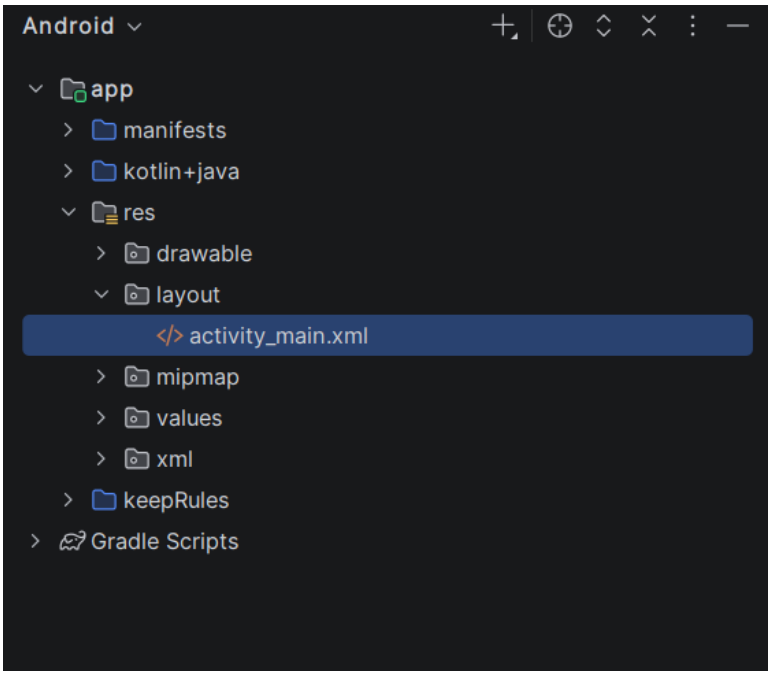
            </intent-filter>

        </activity>

    </application>

</manifest>
```

## activity_main.xml




```xml
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:fitsSystemWindows="true"
    tools:context=".MainActivity">

    <!-- Blue App Bar -->
    <TextView
        android:layout_width="match_parent"
        android:layout_height="56dp"
        android:background="#3F51B5"
        android:gravity="center_vertical"
        android:paddingStart="16dp"
        android:text="GPSTrace"
        android:textColor="@android:color/white"
        android:textSize="20sp" />

    <!-- Main Content -->
    <RelativeLayout
        android:layout_width="match_parent"
        android:layout_height="0dp"
        android:layout_weight="1"
        android:padding="16dp">

        <!-- Latitude -->
```

```
        <TextView
            android:id="@+id/textView"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_alignParentStart="true"
            android:layout_alignParentTop="true"
            android:layout_marginTop="12dp"
            android:text="Latitude = Waiting..."
            android:textSize="20sp" />

        <!-- Longitude -->
        <TextView
            android:id="@+id/textView3"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_below="@id/textView"
            android:layout_alignParentStart="true"
            android:layout_marginTop="95dp"
            android:text="Longitude = Waiting..."
            android:textSize="20sp" />

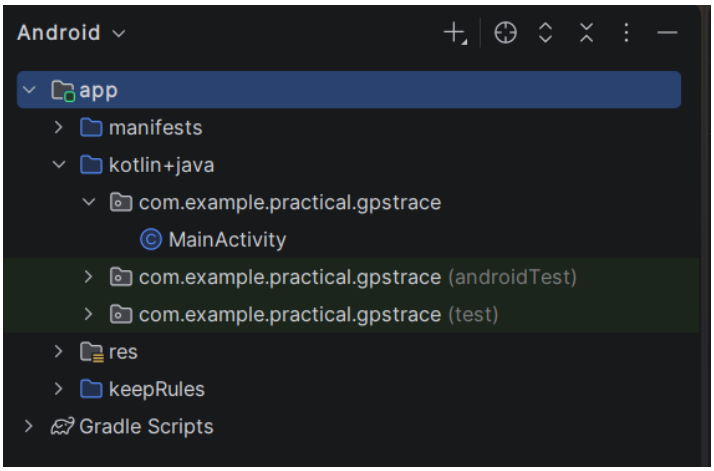
        <!-- Button -->
        <Button
            android:id="@+id/button"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_below="@id/textView3"
            android:layout_centerHorizontal="true"
            android:layout_marginTop="35dp"
            android:text="GET LONGITUDE AND LATITUDE" />

        <!-- College Area Status -->
        <TextView
            android:id="@+id/textView2"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_alignParentBottom="true"
            android:layout_centerHorizontal="true"
            android:layout_marginBottom="12dp"
            android:text="Checking College Area..."
            android:textSize="20sp" />

    </RelativeLayout>

</LinearLayout>
```

# MainActivity.java




```java
package com.example.practical.gpstrace;



import android.Manifest;

import android.content.Context;

import android.content.pm.PackageManager;

import android.location.Location;

import android.location.LocationListener;

import android.location.LocationManager;

import android.os.Bundle;

import android.util.Log;

import android.widget.Button;

import android.widget.TextView;

import android.widget.Toast;
```

```
import androidx.annotation.NonNull;

import androidx.appcompat.app.AppCompatActivity;

import androidx.core.app.ActivityCompat;



public class MainActivity extends AppCompatActivity

       implements LocationListener {



   private static final int LOCATION_PERMISSION_REQUEST_CODE = 100;

    private TextView t1, t2, t3;

    private Button b1;



    private LocationManager locationManager;



    private double lat = 0.0;

    private double longg = 0.0;



    @Override

    protected void onCreate(Bundle savedInstanceState) {
```

```
super.onCreate(savedInstanceState);



setContentView(R.layout.activity_main);




// Connect XML elements

t1 = findViewById(R.id.textView);

t2 = findViewById(R.id.textView3);

t3 = findViewById(R.id.textView2);

b1 = findViewById(R.id.button);




// Get Location Manager

locationManager =

        (LocationManager)

                getSystemService(Context.LOCATION_SERVICE);




// Button click
```

```
        b1.setOnClickListener(view -> updateUI());




        // Check location permission
        checkAndRequestLocationPermission();
    }
```

```
    // ------------------------------------------------------
    // CHECK LOCATION PERMISSION
    // ------------------------------------------------------

    private void checkAndRequestLocationPermission() {

        if (ActivityCompat.checkSelfPermission(
                this,
                Manifest.permission.ACCESS_FINE_LOCATION
        ) != PackageManager.PERMISSION_GRANTED) {

            ActivityCompat.requestPermissions(
```

```
                this,
                new String[]{
                        Manifest.permission.ACCESS_FINE_LOCATION,
                        Manifest.permission.ACCESS_COARSE_LOCATION
                },
                LOCATION_PERMISSION_REQUEST_CODE
        );
```

```
    } else {
        startLocationUpdates();
    }
}

// ----------------------------------------------------------
// START LOCATION UPDATES
// ----------------------------------------------------------

private void startLocationUpdates() {
```

```
try {



    if (ActivityCompat.checkSelfPermission(
            this,
            Manifest.permission.ACCESS_FINE_LOCATION
    ) != PackageManager.PERMISSION_GRANTED) {



        return;
    }



    // Get last known GPS location
    Location lastGps =
            locationManager.getLastKnownLocation(
                    LocationManager.GPS_PROVIDER
            );



    // Get last known Network location
```

```
        Location lastNetwork =

                locationManager.getLastKnownLocation(

                        LocationManager.NETWORK_PROVIDER

                );







        // Use GPS location if available

        if (lastGps != null) {
```

```
            lat = lastGps.getLatitude();

            longg = lastGps.getLongitude();



        }



        // Otherwise use Network location

        else if (lastNetwork != null) {



            lat = lastNetwork.getLatitude();

            longg = lastNetwork.getLongitude();

        }
```

```
// Request GPS location updates
if (locationManager.isProviderEnabled(
        LocationManager.GPS_PROVIDER)) {

    locationManager.requestLocationUpdates(
            LocationManager.GPS_PROVIDER,
            2000,
            1,
            this
    );
}

// Request Network location updates
if (locationManager.isProviderEnabled(
        LocationManager.NETWORK_PROVIDER)) {

    locationManager.requestLocationUpdates(
```

```
                    LocationManager.NETWORK_PROVIDER,

                    2000,

                    1,

                    this

            );

        }
```

```
    } catch (Exception e) {

        Log.e(

                "GPS_TRACE",

                "Error starting location updates: "

                        + e.getMessage()

        );

    }

}



// ------------------------------------------------------
```

```
// UPDATE SCREEN

// ------------------------------------------------------------



private void updateUI() {



    // Display latitude

    t1.setText(

            "Latitude = " + lat

    );




    // Display longitude

    t2.setText(

            "Longitude = " + longg

    );




    // Check whether device is inside college area

    evaluateGeofence(lat, longg);

}
```

```
// ------------------------------------------------------------

// COLLEGE GEOFENCE

// ------------------------------------------------------------



private void evaluateGeofence(

        double currentLat,

        double currentLong) {



    // Exact college coordinates

    double collegeLat = 19.205933;

    double collegeLong = 72.873450;



    // College area radius

    // 200 meters around the college

    float collegeRadius = 200;
```

```
// Create Location object for college
Location collegeLocation =
        new Location("College");

collegeLocation.setLatitude(collegeLat);
collegeLocation.setLongitude(collegeLong);
```

```
// Create Location object for current device
Location currentLocation =
        new Location("Current Location");

currentLocation.setLatitude(currentLat);
currentLocation.setLongitude(currentLong);


// Calculate distance between device and college
float distance =
        currentLocation.distanceTo(
```

```
                    collegeLocation

            );




    // Check whether device is inside college area

    if (distance <= collegeRadius) {



        t3.setText(

                "Harshita you are inside the College Area :)"

        );



    } else {



        t3.setText(

                "Harshita you are outside the College Area :("

        );

    }




    // Show distance in Logcat
```

```
        Log.d(

                "GPS_TRACE",

                "Distance from college = "

                        + distance

                        + " meters"

        );

    }
```

```
    // ----------------------------------------------------------

    // LOCATION CHANGED

    // ----------------------------------------------------------


    @Override

    public void onLocationChanged(

            @NonNull Location location) {


        // Get current latitude

        lat = location.getLatitude();
```

```
        // Get current longitude
        longg = location.getLongitude();




        Log.d(
                "GPS_TRACE",
                "Latitude = "
                        + lat
                        + " | Longitude = "
                        + longg
        );
    }



    // ------------------------------------------------------
    // PERMISSION RESULT
    // ------------------------------------------------------

    @Override
```

```
public void onRequestPermissionsResult(

        int requestCode,

        @NonNull String[] permissions,

        @NonNull int[] grantResults) {



    super.onRequestPermissionsResult(

            requestCode,

            permissions,

            grantResults

    );



    if (requestCode ==

            LOCATION_PERMISSION_REQUEST_CODE) {



        if (grantResults.length > 0

                && grantResults[0]

                == PackageManager.PERMISSION_GRANTED) {
```

```
                startLocationUpdates();



            } else {



                Toast.makeText(

                        this,

                        "Location permission is required.",

                        Toast.LENGTH_SHORT

                ).show();

            }

        }

    }
```

```
    // ------------------------------------------------------

    // STOP LOCATION UPDATES

    // ------------------------------------------------------


    @Override

    protected void onPause() {
```

```
    super.onPause();




    if (locationManager != null) {



        if (ActivityCompat.checkSelfPermission(

                this,

                Manifest.permission.ACCESS_FINE_LOCATION

        ) == PackageManager.PERMISSION_GRANTED) {



            locationManager.removeUpdates(this);

        }

    }

}




// ---------------------------------------------------------

// RESTART LOCATION UPDATES

// ---------------------------------------------------------
```

```
    @Override
    protected void onResume() {

        super.onResume();


        if (locationManager != null) {
            if (ActivityCompat.checkSelfPermission(
                    this,
                    Manifest.permission.ACCESS_FINE_LOCATION
            ) == PackageManager.PERMISSION_GRANTED) {

                startLocationUpdates();
            }
        }
    }
}
```

# Strings.xml




```
<?xml version="1.0" encoding="utf-8"?>



<resources>


    <string name="app_name">gpstrace</string>



</resources>
```

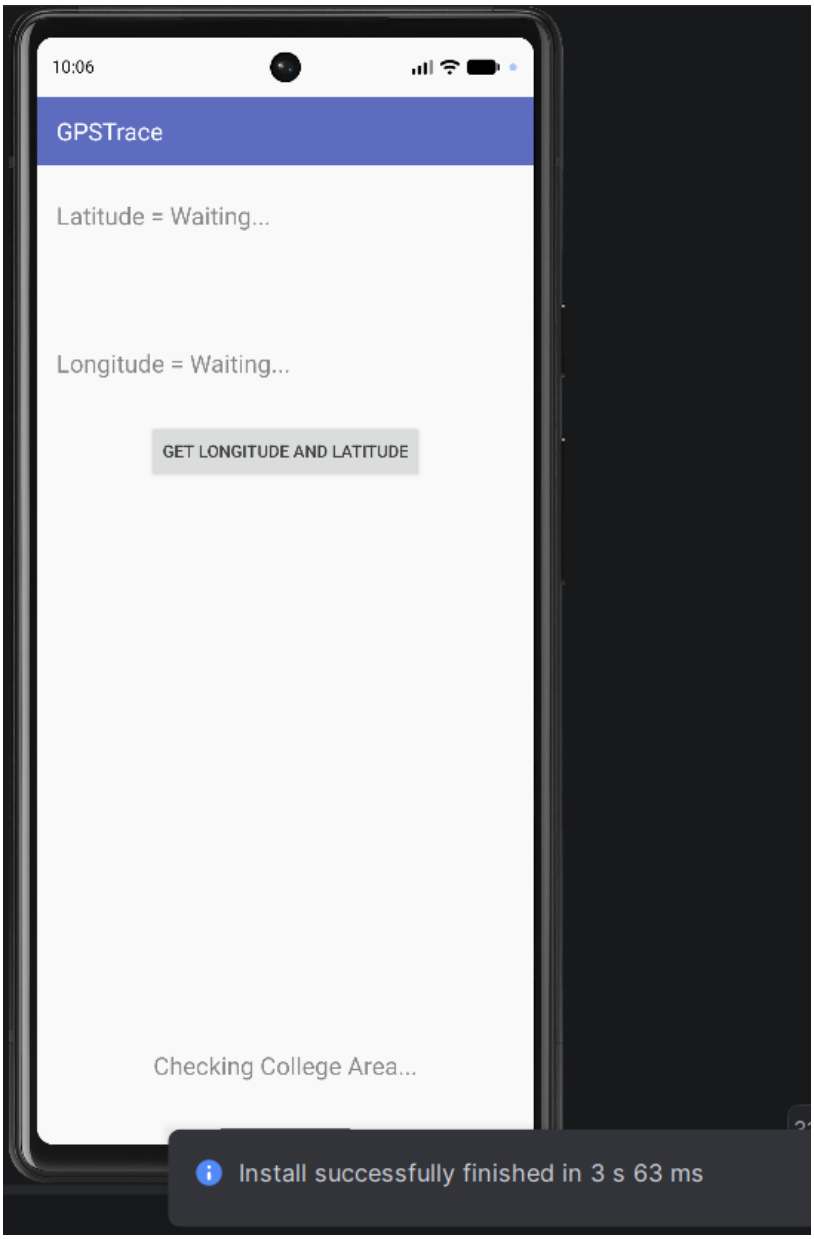
3. **Physical Hardware Deployment (USB Debugging):**
    - Connect the Oppo device to the PC using a USB cable.
    - On the phone, navigate to **Settings > System Settings > Developer Options**.
    - Toggle **USB Debugging** to **ON**.
    - On the prompt *"Allow USB debugging?"*, select **Always allow from this computer** and tap **Allow**.
    - Select the connected device (OPPO CPH1919) from Android Studio's device menu

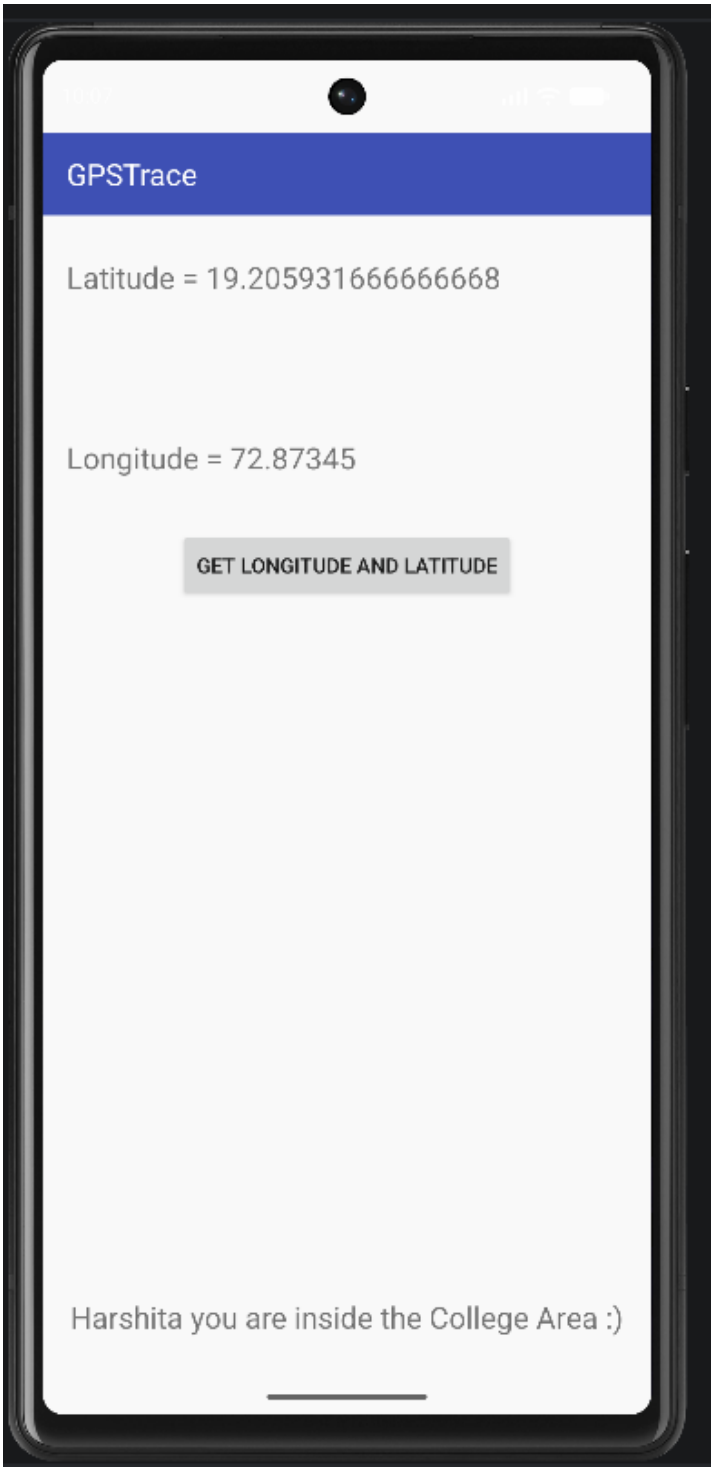
and click **Run (Shift + F10)**.

**Alternate solution:**
Open **Settings**.

- Go to **About phone** / **About device**.-> **version**
- Find **Build number**.
- **Tap "Build number" 7 times** quickly.
- Enter your phone PIN/password if asked.
- You should see a message like **"You are now a developer!"**
- Go back to **Settings** and search for **Developer options**.

**OUTPUT**






**CONCLUSION**

The location-based Android application was successfully designed, compiled, and deployed onto a physical device. Modern runtime permissions and dual location providers were integrated to demonstrate context-aware mobile computing.